FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-	**Team No-**	**Project Title-** Flight Search Engine with Interactive Real-Time Seat Selection System
Team Lead Roll Number		**2520030236**
Team Lead Name		**M JASHWANTH REDDY**
Team Member Roll Number		**2520030291**
Team Member Name		**M RAM RATAN**

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)
This project aims to develop a comprehensive **Flight Search and Interactive Seat Selection System** to streamline the digital travel experience. The proposed solution addresses the complexity of modern travel by providing a unified platform where users can search for flights using dynamic filters such as source, destination, date, and price. Upon selecting a flight, the system transitions into an interactive, color-coded seat map that allows passengers to view real-time availability across Economy, Business, and First classes. The platform is built using a **React.js** frontend for a responsive user interface and a **Node.js/Express.js** backend for handling API requests and seat-lock logic. Key features include e-ticket generation, dynamic price calculation based on seat class, and a mock payment gateway. By integrating these functionalities, the project offers a hackathon-grade prototype that demonstrates proficiency in full-stack development, RESTful API design, and state management.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	**Assigned Roll Number(s)**	**Signature**
CO1	**2520030291**	
CO2	**2520030291**	
CO3	**2520030236**	
CO4	**2520030291**	
CO5	**2520030236**	
CO6	**2520030236**	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

Module Name	CO Mapping	Technical Details	Responsibility
Architectural Foundations	CO1: Framework Design	Establishing declarative UI structures, component-driven architecture, unidirectional data flow	2520030291
JS & TS Engineering	CO2: Logic & UI Correctness	Defining TypeScript interfaces & generics for Flight and Seat data, ensuring modular code	2520030291
Search Component	CO3: Component Model	Designing search bar & filters as controlled components; managing props for flight results	2520030236
Interactive Seat Map	CO3: Component Composition	Building seat map using component composition; real-time class-based color coding	2520030236
Global State Manager	CO4: State Architecture	Lifting state for itinerary sharing; managing global auth state using Context	2520030291
API Integration Layer	CO4: Async Data Engineering	Creating API layer with Axios; handling race conditions, stale state, skeleton UI for loading	2520030291
Routing & Navigation	CO5: SPA Routing	Dynamic & nested routing (/flight/:id/seats); protected routes for secure navigation	2520030236
Form Engineering	CO5: Form Engineering	Building passenger form with validation pipelines and error handling	2520030236
Testing & Reliability	CO6: Testing Pipelines	Writing unit & integration tests for seat selection and search accuracy	2520030236
Deployment & Build	CO6: CI/CD & Deployment	Managing build (Vite), optimization (tree-shaking), deployment to Vercel/Netlify	2520030236

4. Frameworks / Platforms / Software Used
(List the frameworks, libraries, platforms, or tools you plan to use.)

- **Frontend:** HTML5, CSS3, JavaScript (ES6+), React.js, Tailwind CSS.

- **Backend:** Node.js, Express.js.
- **Database:** MySQL (for persistence) and LocalStorage (for client-state).
- **Libraries:** Axios (HTTP), React Router DOM, React Hook Form.
- **Dev Tools:** VS Code, Postman (API Testing), Git.
- **Deployment:** Localhost / Vercel.

Guide Name and Signature (remarks if any) ______________________________